Dachuan Liu

(734) 210-3745 · dachuan@umich.edu · Homepage: dachuan-liu-repo.github.io

Research interests: Multi-Agent Path Finding (MAPF), Multi-Robot Planning & Coordination, Self-Reconfigurable Modular and Swarm Robotics

Education

University of Michigan Ann Arbor, MI
B.S.E. in Computer Engineering | GPA: 3.91/4.00 *Expected May 2027*
- Relevant coursework: Machine Learning, Robot Kinematics and Dynamics, Self-Driving Cars: Perception and Control

Shanghai Jiao Tong University, UM-SJTU Joint Institute Shanghai, China
B.S.E. in Mechanical Engineering (Dual Degree) | GPA: 3.64/4.00 *Expected Aug 2027*
- Relevant coursework: Design and Manufacturing, Mechanical Laboratory, Differential Equations and Linear Algebra

Publications

[C1] **D. Liu**, S. Zhou, S. Zhao, Z. Ren. "LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions." *Under review*, IEEE International Conference on Robotics and Automation (ICRA), 2027. arXiv:2611.10363.

[J1] **D. Liu**, S. Zhao, Z. Ren. "LSRP*-LA: Scalable and Anytime Multi-Agent Path Finding for Large Agents with Asynchronous Actions." *Under review*, IEEE Robotics and Automation Letters (RA-L), 2026.

Academic Service

- Sub-reviewer, *IEEE Transactions on Intelligent Transportation Systems* (T-ITS), 2026

Research Experience

Robotics, Autonomy, and Planning Lab, Shanghai Jiao Tong University Shanghai, China
Research Assistant | Advisor: Prof. Zhongqiang Ren *Apr 2026 – Present*
- Developed **LSRP***, an anytime C++ solver for MAPF with Asynchronous Actions (MAPF-AA) that uses dominance pruning and restart strategies to keep improving solution quality within the time budget. Scales to 1,000 agents and reduces sum of costs by over 40% compared with LSRP; first-author paper submitted to ICRA 2027.
- Developed **LSRP*-LA**, extending LSRP* to large agents with 1×1 to 3×3 footprints, with a swept-area occupancy model and a rule layer that uses recursive push trees and lane evacuation to improve scalability and solution quality. Scales beyond 800 agents and reduces sum of costs by up to 62% compared with Layered LA-LaCAM.

Swarming Modular Robotics, Robotics Department, University of Michigan Ann Arbor, MI
Undergraduate Researcher (ROB 490) | Advisor: Prof. Steven Ceron *Sep 2026 – Present*
- Built a complete cylindrical robot module for a planned swarm of 20–25 robots; modules are designed to attach to neighbors with a rotating internal magnet and reconfigure by rolling around each other.
- Resolved a drive stall with a dual high-torque motor design, rebuilt the power system around a 7.4 V 2S LiPo with a 5–6 V buck converter, and integrated a photoresistor PCB for light sensing; the magnet mechanism showed no observable wear over 200 fatigue cycles.

Multidisciplinary Design Program, University of Michigan Ann Arbor, MI
Undergraduate Researcher | Advisor: Prof. Paul Green *Jan 2026 – Present*
- Designed driving-simulation experiments in Unreal Engine 5 that vary obstacle type and curve radius, and analyze drivers' visual demand using the visual occlusion method.
- Running participant experiments and data analysis for a manuscript in preparation.

Software Engineering Research Group, Shanghai Jiao Tong University Shanghai, China
Research Assistant | Advisor: Prof. Yan Liu *Feb 2024 – Oct 2024*
- Built the teacher-side platform of an interactive teaching system (Python, HTML/CSS/JS, MySQL) for teacher–student interaction and research data management; the platform received a registered software copyright for our team.

Teaching Experience

EECS 216: Introduction to Signals and Systems, University of Michigan Ann Arbor, MI
Instructional Aide *Sep 2026 – Dec 2026*
- Lead 5 lab sections (30 students each), hold weekly office hours, and grade lab reports and exams.

ECE 216: Introduction to Signals and Systems, UM-SJTU Joint Institute Shanghai, China
Teaching Assistant | Instructor: Prof. Heng Qiao *May 2026 – Aug 2026*

- Taught twice-weekly recitations and weekly office hours; led 3 lab sessions (30 students) and graded assignments, lab reports, and exams.

MATH 286: Differential Equations and Linear Algebra, UM-SJTU Joint Institute Shanghai, China
Teaching Assistant | Instructor: Prof. Horst Hohberger *Sep 2025 – Dec 2025*
- Taught weekly recitations covering linear algebra, ODEs, Fourier series/transforms, and Laplace transforms; graded weekly assignments, four quizzes, and two exams.

Guanzhong Primary School (Volunteer Teaching) Yingshang, Anhui, China
Volunteer Teacher *Aug 2024 – Sep 2024*
- Taught geography and basketball at a rural primary school and organized a school-wide sports meeting.

Selected Projects

Learning Priority Orders for Prioritized SIPP in MAPF-AA (ECE 7606) Shanghai, China
C++, Reinforcement Learning *May 2026 - Aug 2026*
- Trained a Plackett-Luce ranking policy with REINFORCE over per-agent conflict features to order agents for prioritized SIPP in MAPF with asynchronous actions; on held-out four maps (den312d, Room64, Warehouse, Random32) instances, raised success rate at 60 agents from 35% to 65% and the largest solved team from 80 to 140 agents.

Drone Delivery Route Optimization (EECS 281) University of Michigan
C++ *Sep 2025 – Dec 2025*
- Solved TSP-style delivery routing on graphs: Prim's MST for connectivity, branch-and-bound with pruning for exact solutions, and arbitrary insertion + 2-opt for near-optimal solutions under strict runtime limits.

Column-Climbing Robot (ME 350) UM-SJTU Joint Institute
SolidWorks, Raspberry Pi, Python *May 2025 – Aug 2025*
- Designed and fabricated a robot that climbs columns of varying diameter; invented a bicycle-clamp locking mechanism that holds position without power, and built the motor and laser-sensor control loop.

Skills

Programming: C++, Python, C, MATLAB
Tools: PyTorch, Simulink, Git
Hardware: SolidWorks, Arduino, Raspberry Pi
English: TOEFL iBT 5.5/6

Honors and Awards

JI Student Development Scholarship, UM-SJTU Joint Institute 2023 – 2024
Outstanding Student Scholarship (Class B), Shanghai Jiao Tong University 2023 – 2024
Registered Software Copyright (interactive teaching platform, team) Sep 2024
Basic Teaching Assistant Certificate, UM-SJTU Joint Institute Aug 2024